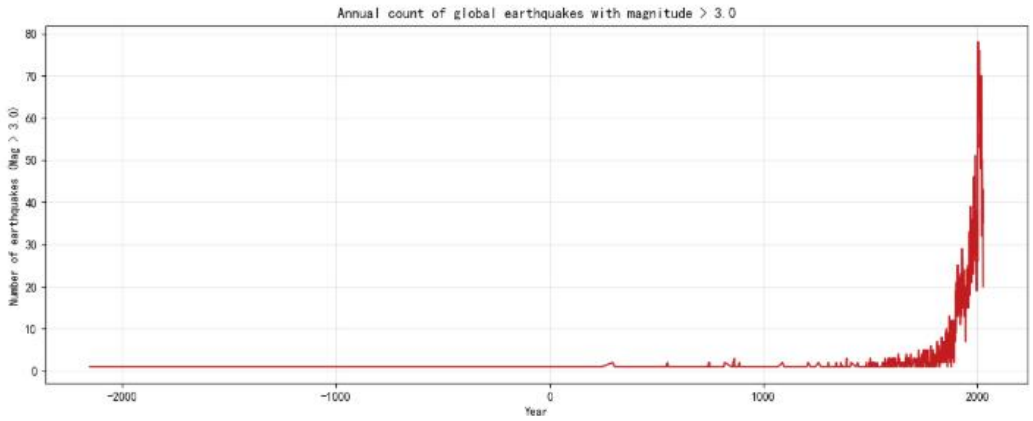
1.Significant earthquakes since 2150 B.C.

1.1

```
===== 1.1 死亡人数前20地区 =====
Location Name
CHINA:  SHAANXI PROVINCE                          830313.0
TURKEY:  ANTAKYA (ANTIOCH)                        434870.0
HAITI:  PORT-AU-PRINCE                            316000.0
IRAN:  TABRIZ                                     268750.0
TURKEY:  ANTAKYA (ANTIOCH), SAMANDAG              250000.0
CHINA:  NE:  TANGSHAN                             242769.0
CHINA:  GANSU PROVINCE, SHANXI PROVINCE           235502.0
AZERBAIJAN:  GYZNDZHA                             230000.0
IRAN:  DAMGHAN, QUMIS                             200000.0
ARMENIA:  DVIN; IRAN:  ARDABIL                    150000.0
CHINA:  GANSU PROVINCE                            137682.0
IRAQ:  AL-MAWSIL                                  120000.0
TURKMENISTAN:  ASHKHABAD                          110000.0
CHINA:  SICHUAN PROVINCE                          107213.0
JAPAN:  TOKYO, YOKOHAMA                           105387.0
CHINA:  NEI MONGOL:  NINGCHENG                    100000.0
PORTUGAL:  LISBON                                  83311.0
AZERBAIJAN:  SHEMAKHA (SAMAXI)                     80000.0
SYRIA:  HALAB (ALEPPO),DIMASHQ (DAMASCUS)          80000.0
ITALY:  MESSINA, SICILY, CALABRIA                  78000.0
Name: Deaths, dtype: float64
```

1.2




```
===== 1.2 每年Mag>3.0地震数量（前10） =====
Year
-2150.0    1
-2000.0    1
-1250.0    1
-1050.0    1
-479.0     1
-426.0     1
-400.0     1
-373.0     1
-330.0     1
-250.0     1
dtype: int64
```

趋势：越靠近现代，记录地震数量明显增加。

原因：不是真实地震频次上升，而是地震监测仪器、台网覆盖逐步完善，早期很多 3 级左右的小地震无法被探测和收录，属于观测记录偏差。

1.3

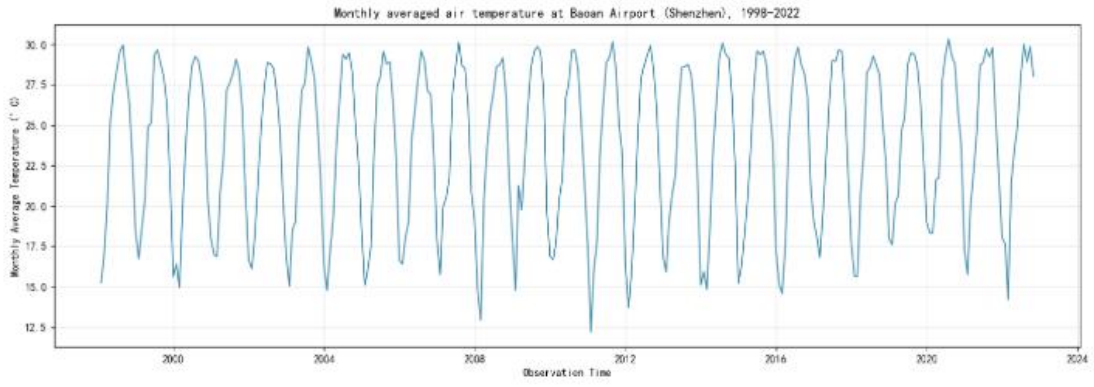
```
===== 1.3 各地地震总数 + 该地最大地震信息（降序） =====
                                Location Name  Total_EQ_count   Largest_EQ_date  \
0                    CHINA:  YUNNAN PROVINCE              68    1833.0-9.0-6.0
1                     RUSSIA:  KURIL ISLANDS              53  1963.0-10.0-13.0
2                   CHINA:  SICHUAN PROVINCE              46   2008.0-5.0-12.0
3                                     TURKEY              45    1944.0-2.0-1.0
4                              VANUATU ISLANDS              40  1913.0-10.0-14.0
...                                        ...             ...              ...
4117        TRINIDAD; GRENADA:  ST GEORGE'S               1   1888.0-1.0-10.0
4118   INDONESIA:  SUMATERA:  BREUEH ISLAND               1   1888.0-3.0-21.0
4119                        URUGUAY:  COLOGNE               1    1888.0-6.0-5.0
4120                      CHINA:  BOHAI GULF                1   1888.0-6.0-13.0
4121  PANAMA: HERRERA, LOS SANTOS, VERAGUAS                1   2026.0-10.0-9.0

                        Largest_EQ_location
0                    CHINA:  YUNNAN PROVINCE
1                     RUSSIA:  KURIL ISLANDS
2                   CHINA:  SICHUAN PROVINCE
3                                     TURKEY
4                              VANUATU ISLANDS
...                                        ...
4117        TRINIDAD; GRENADA:  ST GEORGE'S
4118   INDONESIA:  SUMATERA:  BREUEH ISLAND
4119                        URUGUAY:  COLOGNE
4120                      CHINA:  BOHAI GULF
4121  PANAMA: HERRERA, LOS SANTOS, VERAGUAS

[4122 rows x 4 columns]
```

2. Air temperature in Shenzhen 1998-2022

```
总行数: 235674
清洗后有效记录数: 234877
气温统计(℃):
count    234877.000000
mean         23.768974
std           5.824500
min           1.800000
25%          20.000000
50%          25.000000
75%          28.000000
max          38.000000
Name: Celsius, dtype: float64
```



1998–2022 月平均气温全球尺度存在整体上升趋势。

3. Global collection of hurricanes

3.1

```
===== 3.1 Top 10 storms by WMO_WIND (SID + NAME) =====
                SID       NAME
3433  2015293N13266   PATRICIA
2082  1997253N12255      LINDA
2990  2009288N07267       RICK
1825  1994222N11267       JOHN
104   1959249N19181      PATSY
3453  2016041S14170    WINSTON
2451  2002295N11261      KENNA
656   1979275N06159        TIP
2065  1997212N11265  GUILLERMO
333   1973153N12266        AVA
```

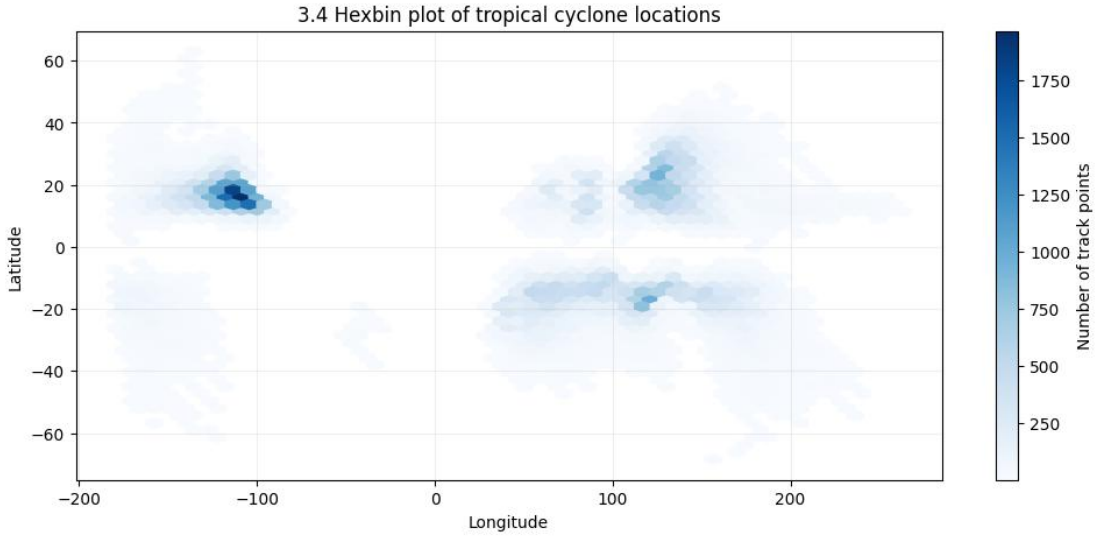
3.2

```
----- 3.2 Top 20 storms wind speed -----
                SID       NAME  WMO_WIND
3433  2015293N13266   PATRICIA     185.0
2082  1997253N12255      LINDA     160.0
2990  2009288N07267       RICK     155.0
1825  1994222N11267       JOHN     150.0
104   1959249N19181      PATSY     150.0
3453  2016041S14170    WINSTON     150.0
2451  2002295N11261      KENNA     145.0
656   1979275N06159        TIP     140.0
2065  1997212N11265  GUILLERMO     140.0
333   1973153N12266        AVA     140.0
3029  2010169N13266      CELIA     140.0
3330  2014234N12261      MARIE     140.0
2441  2002242N14257     HERNAN     140.0
3647  2018226N11245       LANE     140.0
2430  2002204N12263      ELIDA     140.0
2239  1999298N12099        NaN     140.0
3673  2018292N14261      WILLA     140.0
1816  1994202N11238      GILMA     140.0
1813  1994197N09229     EMILIA     140.0
2740  2006228N10218       IOKE     140.0
```

3.3

```
----- 3.3 Count datapoints by Basin -----
BASIN
EP    29736
WP    26235
SI    26144
SP    12458
NI     5293
SA       61
Name: count, dtype: int64
```

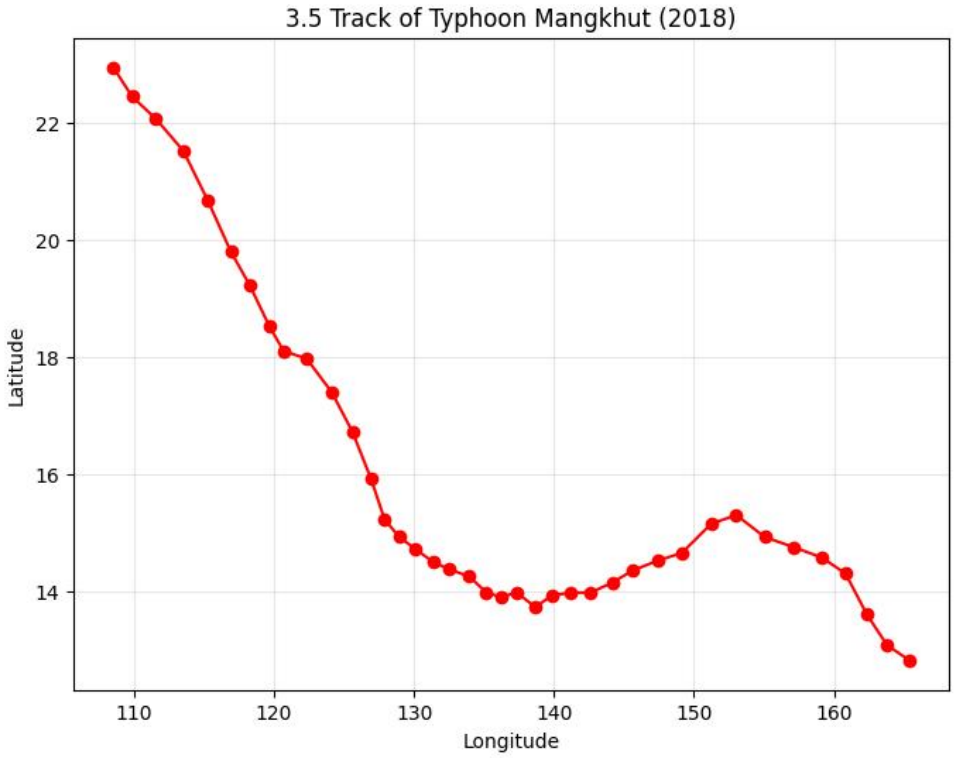
3.4



3.5

----- 3.5 Typhoon Mangkhut (2018) -----

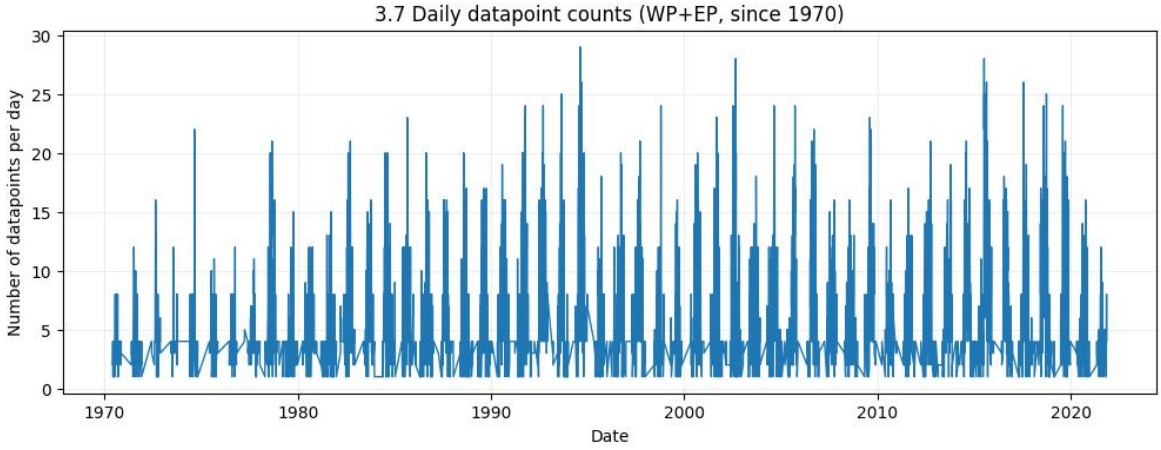
	SID	NAME	ISO_TIME	LAT	LON	WMO_WIND
683026	2018250N12170	MANGKHUT	2018-09-07 12:00:00	12.820	165.367	35.0
683028	2018250N12170	MANGKHUT	2018-09-07 18:00:00	13.075	163.800	35.0
683030	2018250N12170	MANGKHUT	2018-09-08 00:00:00	13.600	162.350	40.0
683032	2018250N12170	MANGKHUT	2018-09-08 06:00:00	14.300	160.825	40.0
683034	2018250N12170	MANGKHUT	2018-09-08 12:00:00	14.575	159.125	45.0
683036	2018250N12170	MANGKHUT	2018-09-08 18:00:00	14.750	157.125	50.0
683038	2018250N12170	MANGKHUT	2018-09-09 00:00:00	14.925	155.050	65.0
683040	2018250N12170	MANGKHUT	2018-09-09 06:00:00	15.300	153.025	65.0
683042	2018250N12170	MANGKHUT	2018-09-09 12:00:00	15.150	151.225	70.0
683044	2018250N12170	MANGKHUT	2018-09-09 18:00:00	14.650	149.125	80.0
683046	2018250N12170	MANGKHUT	2018-09-10 00:00:00	14.525	147.450	80.0
683048	2018250N12170	MANGKHUT	2018-09-10 06:00:00	14.350	145.600	80.0
683050	2018250N12170	MANGKHUT	2018-09-10 12:00:00	14.150	144.200	85.0
683052	2018250N12170	MANGKHUT	2018-09-10 18:00:00	13.975	142.625	90.0
683054	2018250N12170	MANGKHUT	2018-09-11 00:00:00	13.975	141.225	95.0
683056	2018250N12170	MANGKHUT	2018-09-11 06:00:00	13.925	139.825	105.0
683058	2018250N12170	MANGKHUT	2018-09-11 12:00:00	13.725	138.675	110.0
683060	2018250N12170	MANGKHUT	2018-09-11 18:00:00	13.975	137.350	110.0
683062	2018250N12170	MANGKHUT	2018-09-12 00:00:00	13.900	136.225	110.0
683064	2018250N12170	MANGKHUT	2018-09-12 06:00:00	13.975	135.175	110.0
683066	2018250N12170	MANGKHUT	2018-09-12 12:00:00	14.250	133.950	110.0
683068	2018250N12170	MANGKHUT	2018-09-12 18:00:00	14.375	132.500	110.0
683070	2018250N12170	MANGKHUT	2018-09-13 00:00:00	14.500	131.350	110.0
683072	2018250N12170	MANGKHUT	2018-09-13 06:00:00	14.725	130.075	110.0
683074	2018250N12170	MANGKHUT	2018-09-13 12:00:00	14.925	128.950	110.0
683076	2018250N12170	MANGKHUT	2018-09-13 18:00:00	15.225	127.900	110.0
683078	2018250N12170	MANGKHUT	2018-09-14 00:00:00	15.925	126.925	110.0
683080	2018250N12170	MANGKHUT	2018-09-14 06:00:00	16.725	125.600	110.0
683082	2018250N12170	MANGKHUT	2018-09-14 12:00:00	17.400	124.125	110.0
683084	2018250N12170	MANGKHUT	2018-09-14 18:00:00	17.975	122.325	110.0
683086	2018250N12170	MANGKHUT	2018-09-15 00:00:00	18.100	120.700	90.0
683088	2018250N12170	MANGKHUT	2018-09-15 06:00:00	18.525	119.675	90.0
683090	2018250N12170	MANGKHUT	2018-09-15 12:00:00	19.225	118.275	85.0
683092	2018250N12170	MANGKHUT	2018-09-15 18:00:00	19.800	116.925	80.0
683094	2018250N12170	MANGKHUT	2018-09-16 00:00:00	20.675	115.275	75.0
683100	2018250N12170	MANGKHUT	2018-09-16 06:00:00	21.525	113.525	75.0
683106	2018250N12170	MANGKHUT	2018-09-16 12:00:00	22.075	111.550	65.0
683108	2018250N12170	MANGKHUT	2018-09-16 18:00:00	22.450	109.875	55.0
683110	2018250N12170	MANGKHUT	2018-09-17 00:00:00	22.950	108.525	45.0



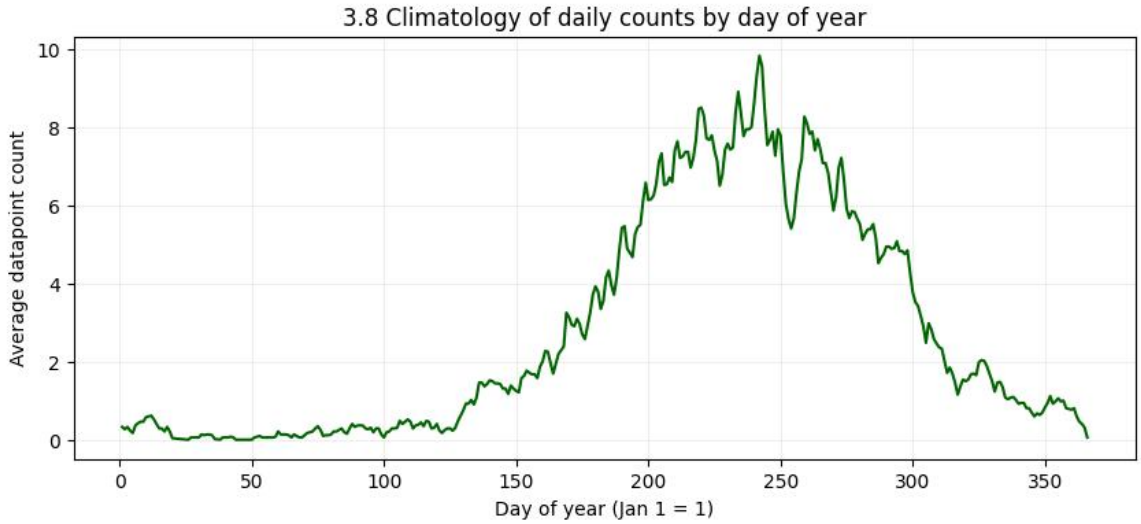
3.6

```
------ 3.6 Subset (>=1970, WP+EP) total points: 52023 ------
```

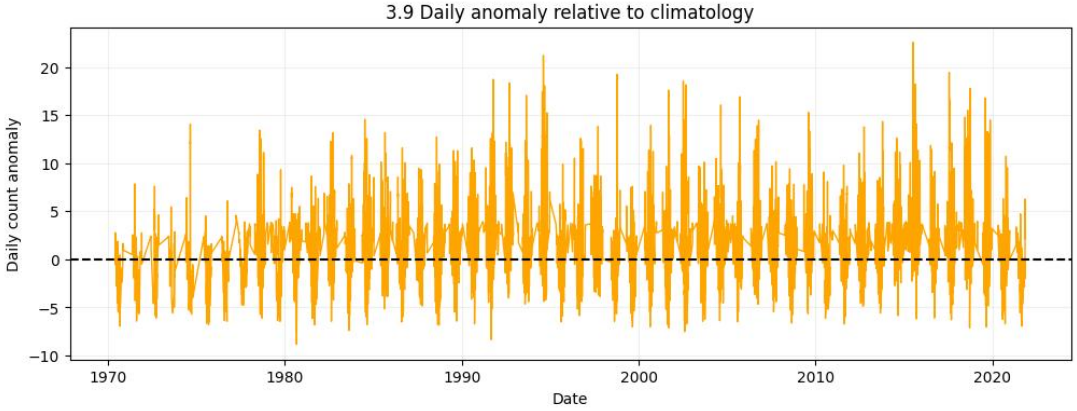
3.7



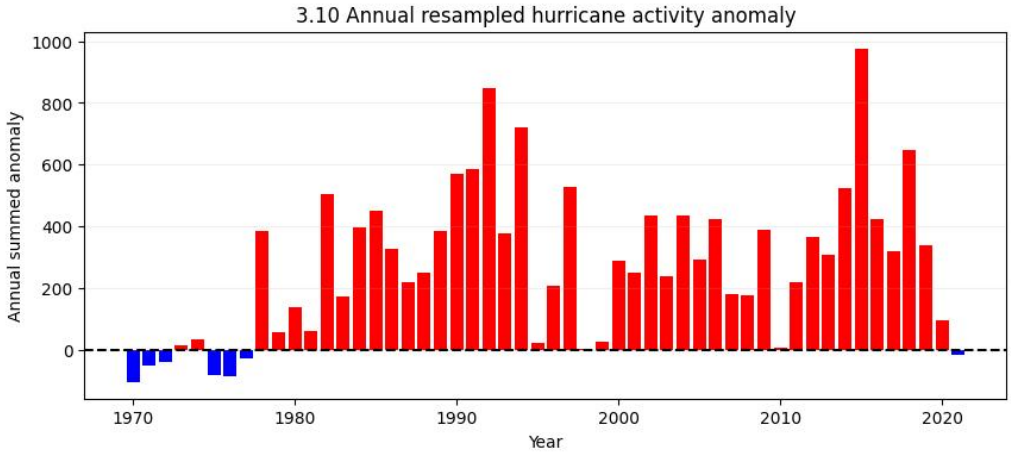
3.8



3.9



3.10



飓风活动显著异常的年份：1992、1994、1997、2015、1970、1975、1976。

4. Explore a data set

选取文件：

Select Data — Total files available based on current filters: 1807

	Compound	Station	Instruments	Data Frequency	Product Type	Info
☑	CFC-11	Adrigole	ALE	High Frequency	Mole Fraction	

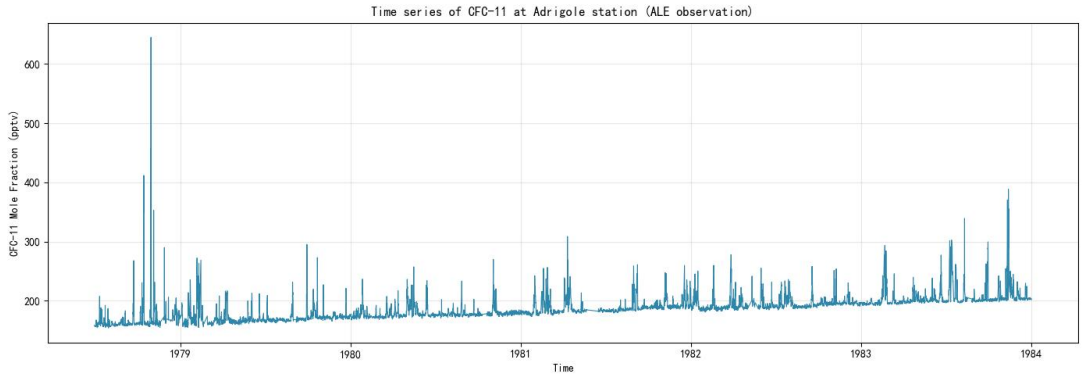
这是爱尔兰 Adrigole 站 ALE 实验观测的 CFC-11（氟利昂-11）大气摩尔分数时间序列数据
时间范围：1978-07-01~1983-12-31

4.1

```
原始数据行数： 9350
清洗后数据行数： 9350

清洗后数据预览：
                 time        mf
0 1978-07-01 00:04:00  159.12134
1 1978-07-01 04:07:00  157.21630
2 1978-07-01 05:08:00  158.11868
3 1978-07-01 06:09:00  157.21630
4 1978-07-03 01:01:00  156.31389
```

4.2



4.3

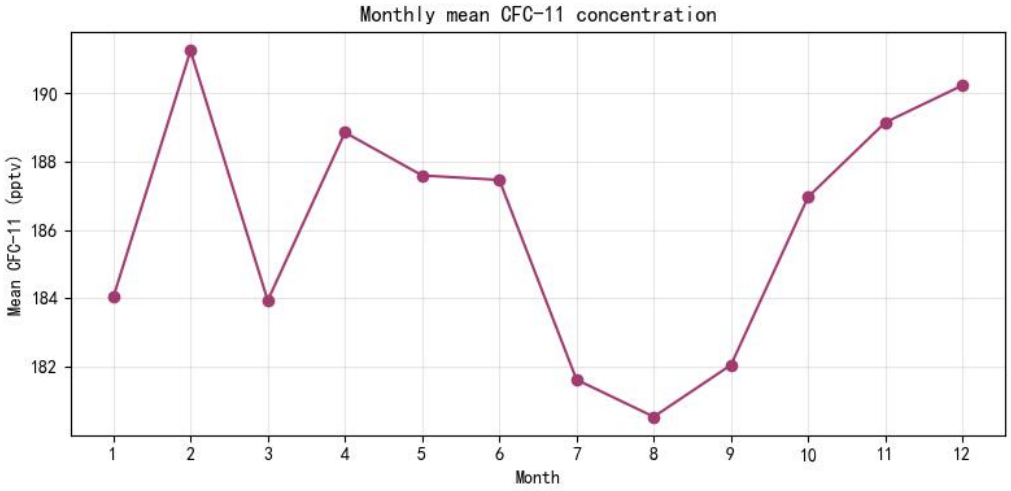
```
----- 【统计检查1：基础描述统计】 -----
均值 = 185.821 pptv
标准差 = 21.903 pptv
中位数 = 183.887 pptv

----- 【统计检查2：长期线性趋势显著性检验】 -----
斜率(每天变化量) = 0.0222 pptv/day
R² = 0.3795
P值 = 0.00e+00
结论：长期趋势统计显著 (p<0.05)
```

均值代表该站点观测时段内 CFC-11 的平均本底浓度；标准差反映观测数据整体离散波动程度；中位数可以用来判断序列是否存在极端高值带来的偏移。
个别异常高观测，极端高值对整体平均有一定拉高作用。
1978-1983 年 CFC-11 整体呈现缓慢上升趋势。

```
----- 【统计检查3：季节变化分析（按月分组平均）】 -----
各月平均CFC-11浓度：
month
1     184.041622
2     191.247069
3     183.943146
4     188.856176
5     187.598658
6     187.467797
7     181.628000
8     180.543537
9     182.056514
10    186.972403
11    189.149078
12    190.233820
Name: mf, dtype: float64
```

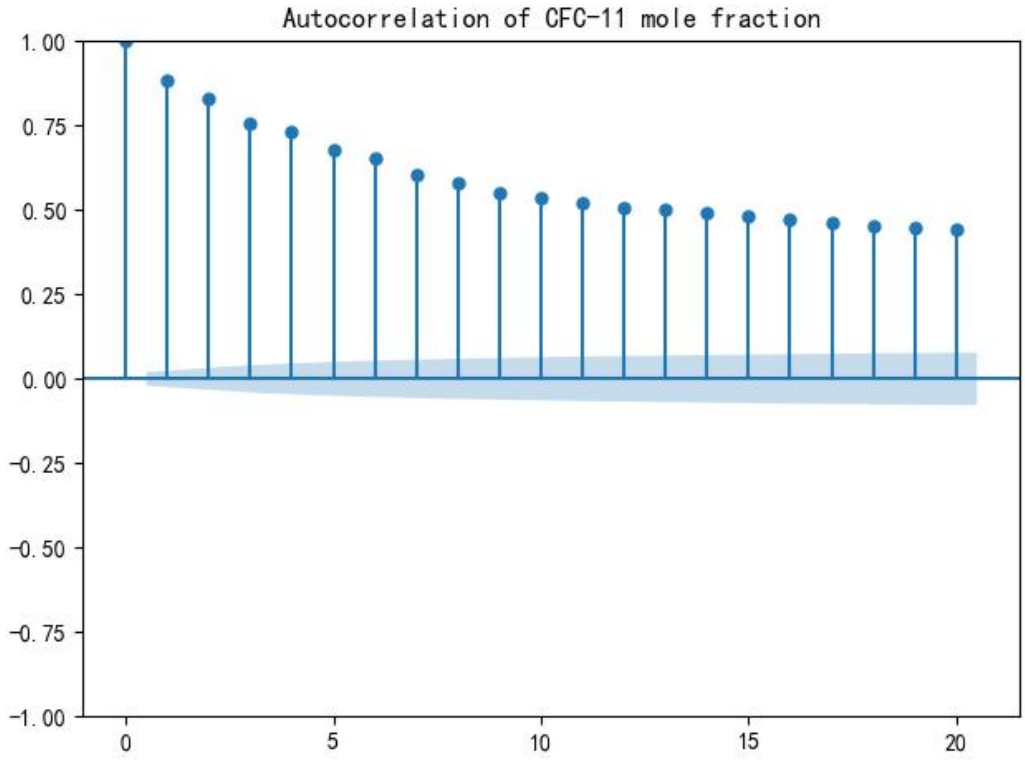
CFC-11 存在明显的月尺度波动，2 月平均浓度最高，8 月平均浓度最低，表现出一定的季节循环特征。




```
----- 【统计检查4：3σ异常值检测】 -----
阈值下限 120.110, 阈值上限 251.531
识别异常样本数量：117
异常样本前5条：
                    time        mf
740 1978-09-22 23:06:00  268.00964
741 1978-09-23 00:07:00  266.70620
939 1978-10-14 04:10:00  351.63110
940 1978-10-14 05:11:00  347.41992
941 1978-10-14 06:12:00  351.33026
```

异常值大概率是仪器瞬时干扰、采样污染，并非真实大气本底浓度。这部分异常样本数量不多，属于少数离群点，不代表长期本底水平。

```
----- 【统计检查5：自相关检验（时序持续性）】 -----
<Figure size 800x400 with 0 Axes>
```



自相关图解读：若前几阶落在置信区间外，说明观测存在时间相关性

滞后 1-20 阶自相关系数全部显著大于 0，落在置信区间之外。

CFC-11 时间序列具有很强的时间自相关性，相邻时刻观测不是相互独立的。大气中的 CFC-11 是缓慢变化的本底示踪气体，前后观测存在记忆效应；做后续统计分析时，不能直接假设样本相互独立。